Reward-Induced Topological Capture in Competitive Graph Plasticity Systems

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Abstract

We study multi-agent systems in which agents with heterogeneous goals competitively modify a shared weighted graph through Hebbian reinforcement proportional to path quality. We report two main findings. First, structural inequality is non-monotonic in the decay rate λ, peaking sharply at $\lambda \approx 0.3$—a *selective consolidation regime* in which forgetting prunes low-reinforcement paths while high-reinforcement dominant paths survive. Second, in a controlled experiment that holds the scoring policy fixed and varies only goal position ($N = 20$, 12 seeds), we do not detect a difference in peak inequality between homogeneous- and heterogeneous-policy agents (peak Gini 0.359 vs. 0.360)—a null result we report under acknowledged limited statistical power. Together, these suggest that the dominant driver of inequality is the interaction between goal-position heterogeneity and selective forgetting, rather than scoring-policy asymmetry: because the reward metric (path coherence) is calibrated on the current graph, it preferentially rewards agents whose goals align with existing topological centrality. The effect is also initialization-robust—granting the weaker agent 50 epochs of sole graph access widens rather than closes the structural gap. We term this mechanism *reward-induced topological capture* and formalize the system as **CRAMPS** (Competitive Reinforcement for Attractor-biased Multi-agent Plasticity in Shared substrates).

1 Introduction

A central problem in multi-agent systems is how local interaction rules produce global structural properties. Prior work on competitive dynamics has examined reward shaping, policy gradient methods [Lowe et al., 2017], and emergent communication [Mordatch and Abbeel, 2018]. Less attention has been paid to how agents sharing a *modifiable environment*—rather than competing for a fixed resource—generate asymmetric structural outcomes from symmetric starting conditions.

We address this gap through competitive graph plasticity. Cognitive substrates, shared knowledge systems, and semantic memory can be modeled as graphs where nodes represent conceptual units and edges encode associative strength [Collins and Loftus, 1975, Steyvers and Tenenbaum, 2005]. When multiple agents navigate such a graph and strengthen traversed edges (a computational analog of Hebbian consolidation [Hebb, 1949]), the resulting dynamics are not merely additive. The differential reinforcement accumulates into structural inequality not through explicit competition, but as a consequence of *goal-position* in the graph under a selective decay regime.

A conventional assumption in cumulative advantage models (e.g., preferential attachment [Barabási and Albert, 1999]) is that monopolization is driven by path dependency or first-mover advantage. We invalidate this assumption for graph plasticity systems. More critically, we show that scoring policy asymmetry is *not* the primary driver of topological capture: agents with identical scoring functions but different positional goals do not produce detectably different structural inequality to agents with heterogeneous policies. Inequality peaks at an intermediate decay $\lambda \approx 0.3$, regardless of policy homogeneity, suggesting that selective forgetting—not policy bias—is the operative mechanism. Agents whose goals align with existing topological centrality systematically accumulate structural advantage because the reward metric (path coherence) is calibrated on the current graph, which already favors central regions. This is the computational analog of meritocracy's structural critique [Barocas et al., 2019]: identical rules do not produce equal outcomes when positional heterogeneity interacts with a structure-dependent evaluation criterion.

Contributions.

1. **Formalization.** We introduce **CRAMPS** (Competitive Reinforcement for Attractor-biased Multi-agent Plasticity in Shared substrates), a multi-agent dynamic system with competitive Hebbian reinforcement proportional to path coherence, operating on a shared weighted semantic graph with selective decay.
2. **Phase transition.** On neutral symmetric graphs ($N = 20$, 315 runs), we identify a non-monotonic Gini curve with maximum at $\lambda \approx 0.3$, the selective consolidation regime.
3. **Null-result kill test (limited power).** A controlled homogeneous-scoring experiment (identical $\alpha, \beta, \gamma, \delta$ for all agents, differing only in goal position; $N = 20$, 12 seeds) does not detect a difference in peak Gini from the heterogeneous-policy condition ($\Delta\,\text{Gini} = 0.001$). We report this as a null result under acknowledged limited statistical power; it is consistent with—but does not prove—the hypothesis that goal-position heterogeneity under selective forgetting is the dominant causal driver.
4. **Initialization invariance.** Counterfactual experiments ($K \in \{10, 50\}$) demonstrate that dominance is independent of initialization order and is amplified by pre-colonization.

2 Related Work

Multi-agent dynamics in modifiable environments. Nash equilibria [Nash, 1950] and price of anarchy [Roughgarden, 2005] study outcomes in fixed-resource games. Multi-agent reinforcement learning [Lowe et al., 2017, Oroojlooy and Hajinezhad, 2023] addresses shared objectives in dynamic settings but not competitive substrate modification. Our work extends these frameworks to environments where agents modify the state space itself.

Dynamic graph systems. EvolveGCN [Pareja et al., 2020] and temporal graph networks [Rossi et al., 2020] model time-varying graphs under centralized optimization. Competitive modification by heterogeneous agents without shared objectives is not addressed by these approaches.

Preferential attachment. Barabási and Albert [Barabási and Albert, 1999] explain power-law degree distributions through age-dependent node advantage. Our mechanism is orthogonal: on a symmetric graph where all nodes have identical initial conditions, scoring bias—not node age—determines structural dominance.

Hebbian plasticity. Spike-timing dependent plasticity [Bi and Poo, 1998] and connectionist models [Rumelhart et al., 1986] apply Hebbian rules at the neural level. We apply them at the semantic graph level, where reinforcement is driven by path coherence.

Structural fairness in algorithms. Barocas et al. [2019] and Dwork et al. [2012] examine how optimization criteria produce discriminatory outcomes without discriminatory intent. Our result is structurally analogous but at the level of graph topology rather than classification decisions.

3 Formal Setup

CRAMPS (Competitive Reinforcement for Attractor-biased Multi-agent Plasticity in Shared substrates) is defined by four components: a shared weighted semantic graph G, a set of heterogeneous agents with scoring functions f_a, a Hebbian plasticity rule proportional to path quality, and an exponential decay mechanism governed by rate λ.

3.1 Graph

Let $G = (V, E, W, \Phi)$ be an undirected weighted graph where $V = \{v_1, \ldots, v_n\}$ are nodes with semantic embeddings $\phi_i \in \mathbb{R}^d$, $E \subseteq V \times V$ contains edge (i, j) iff $\cos(\phi_i, \phi_j) > \theta_{\text{sim}}$, and $W : E \to \mathbb{R}_{>0}$ assigns edge weights initialized from cosine similarity.

3.2 Agent Scoring Function

Agent $a \in \mathcal{A}$ selects next nodes via:

$$f_a(j \mid i, G) \;=\; \alpha_a \cdot \text{sem}(i, j) + \beta_a \cdot \text{attr}(j, G) + \gamma_a \cdot \text{need}(j, g_a) + \delta_a \cdot H(j, G) \tag{1}$$

where $\text{sem}(i, j) = \cos(\phi_i, \phi_j)$, $\text{attr}(j, G) = \text{PageRank}_G(j)$, $\text{need}(j, g_a) = \cos(\phi_j, g_a)$ for goal vector $g_a \in \mathbb{R}^d$, and $H(j, G)$ is the normalized entropy of j's neighborhood weight distribution. The tuple $(\alpha_a, \beta_a, \gamma_a, \delta_a)$ constitutes the agent's *scoring bias*.

3.3 Trajectory Sampling

Trajectory $\tau_a = (v_1, \ldots, v_L)$ is sampled step-by-step:

$$P(v_{t+1} = j \mid v_t, G) = \text{softmax}_{T_a}\big[f_a(j \mid v_t, G)\big] \quad \text{for } j \in \mathcal{N}_k(v_t) \tag{2}$$

where $\mathcal{N}_k(v_t)$ is the top-k neighborhood by edge weight and T_a is agent temperature.

3.4 Competitive Hebbian Plasticity

After trajectory τ_a with path quality $r(\tau_a)$, traversed edges are updated:

$$w_{ij}(t+1) = \text{clip}\big(w_{ij}(t) + \eta \cdot r(\tau_a) \cdot \cos(\phi_i, \phi_j),\ 0,\ W_{\max}\big) \quad \forall (i, j) \in \tau_a \tag{3}$$

Path quality $r(\tau_a)$ is topic coherence: mean cosine similarity of path embeddings to the path centroid. This implements *competitive* reinforcement: agents producing more coherent paths reinforce more strongly.

All edges decay per epoch:

$$w_{ij}(t+1) \leftarrow w_{ij}(t) \cdot (1 - \lambda) \tag{4}$$

Edges with $w_{ij} < \theta_{\text{prune}}$ are removed.

3.5 Three Experimental Archetypes

Table 1: Agent scoring parameters ($\alpha = 1.0$ for all; α weights local semantic similarity and is held constant throughout).

Agent	α	β	γ	δ	T	Goal
Conformist	1.0	1.2	0.7	−0.5	0.8	corpus centroid
Disruptive	1.0	0.2	0.9	+0.4	1.5	peripheral node
Opportunist	1.0	0.9	0.3	−0.1	1.0	max PageRank

3.6 Structural Inequality Metric

Gini coefficient of edge weight distribution at time t:

$$\text{Gini}(G_t) = \frac{2 \sum_k k \cdot w_{(k)}}{n \sum_k w_{(k)}} - \frac{n+1}{n} \tag{5}$$

where $w_{(k)}$ are edge weights in ascending order. Gini = 0: uniform weights; Gini = 1: single dominant edge.

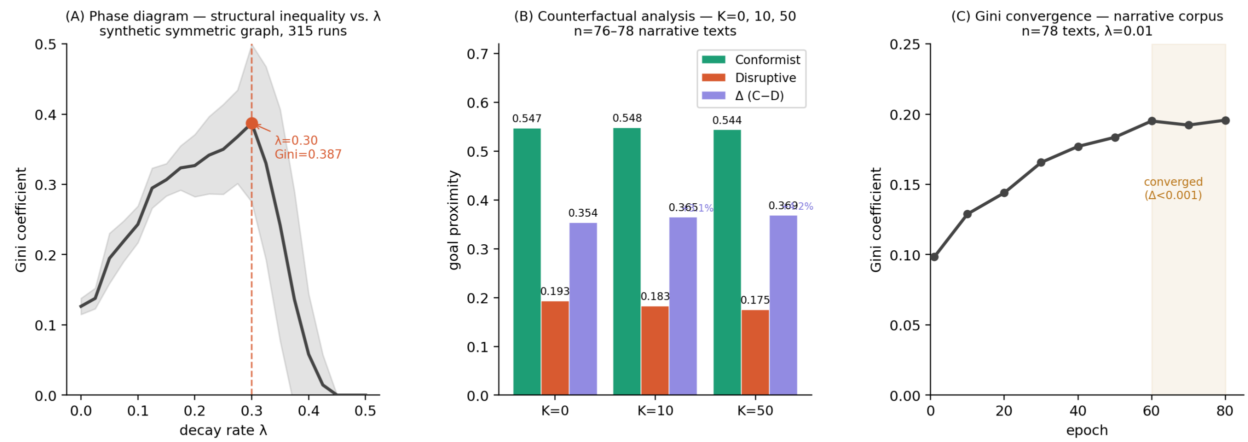

Figure 1: **(A)** Phase diagram: Gini coefficient vs. decay rate λ on neutral symmetric graphs (315 runs). Non-monotonic curve peaks at $\lambda \approx 0.3$ (selective consolidation regime). **(B)** Counterfactual analysis: goal proximity per agent under $K = 0, 10, 50$ head-start conditions. Structural gap $\Delta = (\mathrm{C} - \mathrm{D})$ widens with K. **(C)** Gini convergence on narrative corpus ($n = 78$ texts, $\lambda = 0.01$, 80 epochs). System reaches stable fixed point ($\Delta_{60\to80} < 0.001$).

4 Results

4.1 Phase Analysis: Non-Monotonic Gini as a Function of λ

To isolate scoring bias from corpus structure, we use Watts-Strogatz graphs ($n = 20$, $k = 4$, $p = 0.2$) with uniformly random initial edge weights $W_{ij} \sim \mathcal{U}(0.1, 1.0)$ and random unit-sphere node embeddings ($d = 32$). Two pure agents compete: an attractor agent ($\beta = 1$, all others zero) and an explorer agent ($\delta = 1$, all others zero). We sweep $\lambda \in [0, 0.5]$ across 15 random seeds (315 runs, 80 epochs each, no language embeddings).

Proposition 1 (Phase-Conditioned Topological Capture). *In a competitive graph plasticity system on a neutral symmetric graph, attractor- based scoring achieves structural dominance (73% frequency) only in the memory- preserving regime ($\lambda = 0$). Structural inequality peaks at $\lambda \approx 0.3$ (Gini $= 0.387$). At $\lambda \geq 0.5$, decay destroys structure faster than reinforcement can consolidate it, collapsing inequality to zero.*

Table 2: Phase analysis with *pure agents* (attractor $\beta = 1$ vs. explorer $\delta = 1$): selected λ values (315 runs, neutral symmetric graphs). The peak Gini here (0.387) is from pure agents and is distinct from the heterogeneous-archetype peak (0.360, Table 4).

λ	Gini (mean)	Attractor dominance
0.000	0.126	73%
0.100	0.243	0%
0.200	0.310	22%
0.300	**0.387**	18%
0.400	0.280	10%
0.500	0.000	0%

Three dynamical regimes emerge (Figure 1A):

Memory-preserving ($\lambda \to 0$). All traversal traces persist. Gini is moderate (0.126) because many paths survive; attractor wins 73% by compounding advantage on already-strong edges.

Selective consolidation ($\lambda \approx 0.3$). Decay prunes weak explorer paths but is outpaced by the conformist's repeated reinforcement of dominant edges. *Forgetting acts as an inequality amplifier*, selectively erasing low-reinforcement diversity. Maximum Gini (0.387).

High-decay ($\lambda \to 0.5$). Structure resets each epoch. Explorer dominates by default. Gini $\to 0$.

4.2 Counterfactual: Initialization-Robust Dominance

To test whether attractor dominance stems from path dependency or scoring geometry, we run three conditions on the narrative corpus ($n = 76$–78 texts, 80 epochs): *simultaneous* start; *disruptive head start* of K epochs; *conformist head start* of K epochs. We test $K \in \{10, 50\}$.

Table 3: Goal proximity (post head-start epochs) under counterfactual conditions.

Condition	Conformist	Disruptive	Opportunist	Δ (C−D)
Simultaneous ($K = 0$)	**0.547**	0.193	0.397	0.354
Disruptive HS ($K = 10$)	**0.548**	0.183	0.394	0.365
Disruptive HS ($K = 50$)	**0.544**	0.175	0.391	**0.369**

Proposition 2 (Initialization-Robust Dominance)**.** *Conformist goal proximity is dominant (*≈ 0.547*) across all conditions. Granting the disruptive agent 50 epochs of sole graph access (63% of total epochs) does not mitigate dominance; the structural gap* Δ *increases with* K *(*$0.354 \to 0.369$*, +4.2%,* $n = 40$ *texts, no confidence intervals reported). This is a small absolute effect; replication with increased seed count and error estimation is warranted before strong causal attribution.*

Proposition 3 (Pre-Structured Substrate Paradox)**.** *A graph pre-colonized by the disruptive agent provides a stronger substrate for subsequent attractor dominance than an unstructured graph. Pre-existing topological structure, regardless of which agent created it, preferentially benefits high- coherence scoring strategies.*

4.3 Kill Test: Policy Homogeneity Does Not Prevent Topological Capture

Experimental design. To isolate the causal role of scoring policy asymmetry, we run a controlled *homogeneous-scoring* condition (Condition B): all three agents receive identical scoring weights ($\alpha = 1.0$, $\beta = 0.7$, $\gamma = 0.6$, $\delta = -0.1$, $T = 1.1$), differing only in their goal vectors (centroid, peripheral node, dominant node). Parameters match the main experiment: $N = 20$, Watts-Strogatz, 80 epochs, 12 seeds, full λ sweep.

Result. Table 4 shows Gini at each λ value for both conditions. The peaks occur at different λ values: Condition A (heterogeneous) peaks at $\lambda = 0.30$ with Gini $= 0.360$; Condition B (homogeneous) peaks at $\lambda = 0.25$ with Gini $= 0.359$. Peak-to-peak $\Delta\,$Gini $= 0.001$. We report this as a *null result under limited statistical power* (12 seeds; no confidence intervals reported). Point-wise differences at individual λ values are large and noisy (range: -0.012 to $+0.038$), which confirms that 12 seeds cannot resolve the two conditions at any fixed λ. The appropriate reading is: we do not detect a significant difference with this experimental design, not that no difference exists.

Table 4: Kill test: Gini at each λ value, heterogeneous vs. homogeneous scoring ($N = 20$, 12 seeds).

λ	Gini-A (heterog.)	Gini-B (homog.)	Δ
0.00	0.158	0.159	−0.001
0.10	0.328	0.290	+0.038
0.20	0.327	0.330	−0.003
0.25	0.346	**0.359**	−0.012
0.30	**0.360**	0.333	+0.027
0.40	0.252	0.259	−0.007
0.50	0.234	0.224	+0.011

Proposition 4 (Kill Test: Null Result Under Limited Power)**.** *In a CRAMPS system with $N = 20$, 80 epochs, and 12 seeds, we do not detect a statistically significant difference in peak Gini between heterogeneous-policy agents and homogeneous-policy agents differing only in goal position (peak Gini 0.360 vs. 0.359; $\Delta\,\mathrm{Gini} = 0.001$). This is a null result under acknowledged limited statistical power: absence of a detected difference does not constitute evidence of equivalence. The result is consistent with the hypothesis that goal-position heterogeneity under selective forgetting is the dominant causal driver, but higher-powered replication is required to confirm it.*

Interpretation. The agent whose goal aligns with the topological centroid of the graph systematically produces more coherent paths—not because its scoring function is superior, but because coherence (the reward metric) is calibrated on the current graph, which already concentrates edges near the centroid. The reward metric itself is not neutral: it preferentially assigns higher values to paths through already-dense regions, regardless of whether the agent was designed to prefer those regions. Selective forgetting at $\lambda \approx 0.3$ then amplifies this positional advantage by pruning peripheral paths.

4.4 Convergence: Stable Gini Fixed Point

On the narrative corpus ($n = 78$, $\lambda = 0.01$, 80 epochs), Gini rises from 0.098 to 0.196 and stabilizes ($\Delta_{\text{epochs } 61\text{–}80} = -0.00025$), consistent with the phase analysis at low λ (Figure 1C).

5 Discussion

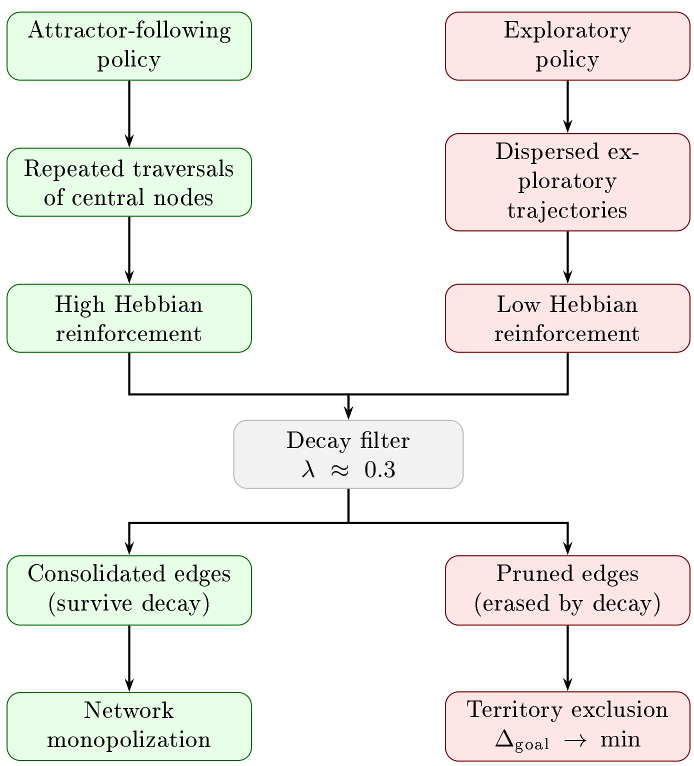

Figure 2: Schematic mechanism of *reward-induced topological capture* at the selective consolidation regime ($\lambda \approx 0.3$). Decay acts as a selective filter: high-reinforcement paths survive; low-reinforcement exploratory paths are pruned, producing structural monopolization without explicit competitive intent. This diagram is conceptual; quantitative peak values appear in Tables 2 (pure agents, Gini = 0.387) and 4 (heterogeneous agents, Gini = 0.360).

Forgetting as inequality amplifier. The $\lambda \approx 0.3$ peak defines the selective consolidation regime where decay acts as a selection pressure: it eliminates low-reinforcement paths faster than high-reinforcement paths. The mechanism is analogous to natural selection operating on path strength rather than biological fitness. Partial forgetting *institutionalizes* the monopoly by continuously pruning diversity—and, critically, this process operates independently of whether the dominant agent has a privileged scoring function or merely a privileged goal position.

The metric is not neutral—and a design note on circularity. The kill test (Proposition 4) suggests a mechanism that extends beyond multi-agent RL: *the reward metric inherits the topology of the current graph.* Path coherence—the quality measure used by all agents—is higher near topologically central regions because those regions already contain dense edge structure. An agent pursuing the centroid therefore generates higher-quality paths not because it is better designed, but because quality, as measured by coherence, is structurally biased toward the centroid.

We name a design coupling explicitly: in the main heterogeneous condition (Condition A), the Conformist agent's *goal* is the corpus centroid and the *reward* is topic coherence, which is closeness to the path centroid. This creates a partial tautology—the agent whose goal aligns with the reward reference point will score higher by construction. The claim that this is *emergent* should therefore be hedged: the non-monotonic λ phase diagram and the initialization-invariant dominance (Propositions 1 and 2) are the dynamically non-obvious results; the specific conformist advantage in Condition A is partly a consequence of experimental design. The kill test (Condition B) partially addresses this by using heterogeneous goals with a neutral scoring function, but the circularity concern applies there too—the centroid-goal agent still aligns better with coherence. This is a limitation the kill test exposes rather than resolves. This is the computational analog of what Barocas et al. [2019] identify as structural bias: identical rules applied to positionally unequal agents produce unequal outcomes.

Separation from preferential attachment. Preferential attachment [Barabási and Albert, 1999] predicts monopolization from node-age advantage. Our result is orthogonal on two counts: (1) on symmetric graphs with identical initial conditions and identical scoring policies, goal-position heterogeneity alone drives dominance; (2) the pre-structured substrate paradox further separates the mechanisms—a weaker agent colonizing the graph first does not gain protection but rather reduces initial entropy, enabling the stronger-positioned agent to exploit the resulting structure more efficiently.

Implication for system design and alignment (preliminary). The results above are established on Watts-Strogatz graphs with $N = 20$ and a single narrative corpus ($n \approx 78$). The following implications are directional hypotheses consistent with the data, not validated predictions: a shared representational substrate operating at $\lambda \approx 0.3$ may exhibit elevated structural inequality even under formally equal evaluation rules. If the reward criterion is calibrated to optimize coherence over an existing representational space, it preferentially rewards agents whose goals align with existing topological centrality—a form of structural bias described in Barocas et al. [2019]. Validation on larger graphs ($N \in \{50, 100, 200\}$) and multiple corpora is required before drawing implications for real knowledge-graph or AI-alignment systems.

6 Experimental Details

Narrative corpus. 289 raw texts from WritingPrompts (HuggingFace: `euclaise/writingprompts`); after quality filtering (200–1500 chars, ≥ 4 sentences, minimal formatting artifacts, seed 42), $n = 76$–78 texts remain and form the effective corpus for Sections 4.2–4.4.

Synthetic graphs. Watts-Strogatz: $n = 20$, $k = 4$, $p = 0.2$. Edge weights $W_{ij} \sim \mathcal{U}(0.1, 1.0)$; node embeddings: random unit vectors in $\mathbb{R}^{32}$. No language model used—no corpus bias.

Embeddings (narrative). `all-MiniLM-L6-v2` ($d = 384$), L2-normalized.

Graph. Cosine similarity threshold $\theta_{\text{sim}} = 0.35$; top-$k = 10$; pruning $\theta_{\text{prune}} = 0.005$.

Plasticity. $\eta = 0.05$, $W_{\max} = 3.0$, normalization every 5 epochs.

Decay sweep. $\lambda \in \{0.000, 0.025, \ldots, 0.500\}$ (21 values), 15 seeds per value.

Walk. $L = 8$ steps; 5 runs per condition per text.

Counterfactual. $K \in \{0, 10, 50\}$ head-start epochs. Dominant agent: highest mean goal proximity over post-K window.

Implementation. Python 3.12, NetworkX 3.6, `sentence-transformers`, Gudhi 3.12, SciPy 1.17, NumPy 2.4. Code: https://github.com/manolo-delabarra/cramps.

7 Limitations

1. **Empirical propositions.** Propositions 1–4 are empirical. A Lyapunov stability analysis characterizing the fixed point and the $\lambda^* \approx 0.3$ peak in closed form remains open.
2. **Kill test statistical power.** Proposition 4 uses 12 seeds with no confidence intervals; the peaks occur at different λ values (A=0.30, B=0.25); and point-wise differences range from -0.012 to $+0.038$. This is a null result under limited power, not evidence of equivalence. Replication with $n \geq 20$ seeds, confidence intervals, and a formal equivalence test is required.
3. **Metric–goal circularity.** In Condition A, the Conformist's goal is the corpus centroid and the reward is coherence (closeness to path centroid). This creates a structural coupling that is partially tautological, not fully emergent. The genuinely non-obvious results are the non-monotonic λ curve and the initialization-invariant dominance, not the Conformist's advantage per se.
4. **Small effects with large labels.** The pre-structured substrate paradox rests on a +4.2% gap increase ($0.354 \to 0.369$) without error bars. This warrants replication before the label "paradox" is applied.
5. **Scale and generalizability.** Results are from $N = 20$ graphs and $n \approx 78$ texts. Claims about real knowledge graphs, memory systems, or AI alignment require validation at $N \in \{50, 100, 200\}$ across multiple corpus domains.
6. **Reward metric scope.** Preliminary experiments suggest that novelty or anti-centrality rewards invert dominance, but a full λ sweep with $n \geq 20$ seeds is needed before this constitutes a formal claim.

8 Conclusion

The most robust finding is a non-obvious one: structural inequality in competitive graph plasticity is non-monotonic in the decay rate λ, peaking at $\lambda \approx 0.3$ (the selective consolidation regime) across 315 synthetic runs. A null-result kill test ($N = 20$, 12 seeds; acknowledged low power) does not detect a difference between heterogeneous- and homogeneous-policy conditions, consistent with the hypothesis that goal-position heterogeneity under selective forgetting—rather than scoring-policy asymmetry—is the dominant causal driver. A structural coupling between the Conformist's goal (corpus centroid) and the reward metric (coherence) partially limits the emergent interpretation; this circularity is named and discussed rather than elided. Dominance is also initialization-robust: granting the weaker agent 50 epochs of sole access amplifies rather than closes the gap (+4.2%, no error bars, replication needed). Taken together, these results establish *reward-induced topological capture* as a mechanism distinct from preferential attachment and demonstrate that structural inequality in shared representational environments is not caused by biased evaluation rules but by the geometry of goal-position under selective forgetting.

A Transdisciplinary Isomorphisms

The formal results in the main paper describe a dynamical system in computational terms. This appendix identifies structural isomorphisms between competitive graph plasticity and two independent bodies of knowledge: neurobiological systems (Section A.1) and social theory (Section A.2). We also derive a formal intervention framework (Section A.3).

We use the term *isomorphism* precisely: we claim that the *formal structure* of the mechanisms is shared, not that the underlying substrates are identical. These mappings generate testable predictions that extend beyond the original simulation.

A.1 Neurobiological Isomorphism

Mapping Table

Table 5: Formal correspondence between competitive graph plasticity and neurobiological mechanisms.

CRAMPS parameter	Neurobiological analog	Reference
Path reward $r(\tau_a)$	Dopaminergic prediction error signal (phasic firing from VTA to nucleus accumbens)	Schultz et al. [1997]
Hebbian update $w_{ij} \mathrel{+}= \eta \cdot r(\tau_a)$	Long-term potentiation (LTP) via AMPA receptor phosphorylation	Bi and Poo [1998]
Decay $w_{ij} \mathrel{\times}= (1-\lambda)$	Synaptic pruning and LTD mediated by microglial activation under cortisol	Sapolsky [2000]
Decay rate λ	Cortisol level (glucocorticoid concentration)	Lupien et al. [2009]
Selective consolidation regime ($\lambda \approx 0.3$)	Eustress optimum in the Yerkes-Dodson curve: intermediate arousal maximizes selective memory consolidation	Yerkes and Dodson [1908]
Gini(G_t)	Synaptic weight distribution inequality in a local circuit	—
Agent temperature T_a	Stochasticity of dopaminergic firing / behavioral variability	Schultz et al. [1997]

The Yerkes-Dodson Curve as Phase Diagram

The inverted-U relationship between cortisol (arousal) and cognitive performance [Yerkes and Dodson, 1908, Lupien et al., 2009] is formally equivalent to the Gini vs. λ phase diagram in Figure 1A. At low cortisol ($\lambda \to 0$), all synaptic traces persist and the network retains structural diversity (Gini = 0.126). At high chronic cortisol ($\lambda \to 0.5$), pruning exceeds consolidation and the network resets to near-homogeneity. At intermediate cortisol ($\lambda \approx 0.3$), *selective consolidation* operates: high-dopamine pathways survive pruning while low-dopamine exploratory pathways are eliminated. This is the neurobiological mechanism producing maximum Gini.

Pathological Regimes

Addiction. Addictive substances produce artificial supraphysiological dopamine release, generating extreme $r(\tau)$ values for drug-associated pathways [Volkow et al., 2016]. Simultaneously, withdrawal stress chronically elevates cortisol ($\lambda \uparrow$), accelerating pruning of non-drug pathways. The result is a neural graph with Gini $\to 1$: one or two dominant circuits (drug-seeking) and catastrophic pruning of alternatives. The pre-structured substrate paradox applies directly: voluntary attempts to activate alternative pathways traverse connections already weakened by cortisol, which the dominant circuit absorbs as context for further strengthening.

Post-Traumatic Stress. Traumatic events combine high salience (high $r(\tau)$) with extreme cortisol (λ temporarily at high-decay regime), producing hyperstable consolidation of threat-associated nodes [Ross et al., 2017]. The resulting graph has anomalously high PageRank for threat nodes; environmental cues activate these via high-weight edges while alternative interpretations remain pruned. Exposure therapy can be formalized as a process of deliberately lowering λ (via safety conditioning) to allow alternative pathways to accumulate weight and compete topologically.

A.2 Social-Theoretical Isomorphism

Gramscian Hegemony

Gramsci's concept of hegemony [Gramsci, 1971] proposes that dominant classes maintain control not primarily through coercion but by ensuring that their evaluative frameworks are accepted as *common sense*. The formal mechanism was left underspecified.

Proposition 2 (Initialization-Robust Dominance) provides a computational instance of the hegemonic mechanism: the conformist agent achieves structural dominance through scoring function geometry, not through prior occupancy, explicit intent, or material advantage. The agent does not *know* it is dominating; it locally optimizes $f_a(j|i, G)$ with $\beta_a = 1.2$, and monopolization emerges as a global consequence.

The selective consolidation regime ($\lambda \approx 0.3$) formalizes what Gramsci called the reproduction of common sense: not total suppression of alternatives (which would correspond to $\lambda \to 0.5$, structural collapse), but a forgetting rate sufficient to degrade counter-hegemonic trajectories while preserving dominant ones. The critical λ is the pace of cultural modernity: fast enough to erode alternative traditions across generations, slow enough to consolidate dominant narrative structures.

Proposition 3 (Pre-Structured Substrate Paradox) formalizes Gramsci's *passive revolution* [Gramsci, 1971]: the process by which counter-hegemonic movements inadvertently stabilize the dominant system. The disruptive agent's colonization reduces graph entropy, creating legible structure that the conformist exploits more efficiently than it could from a flat substrate. Historical instances (labor organizing $\to$ welfare state $\to$ stabilized consumption demand; open-source software $\to$ cloud platform capture) follow this formal pattern.

Bourdieu's Field Theory

Bourdieu's concept of *habitus* [Bourdieu, 1977] describes dispositional structures that reproduce the social field without agents being conscious of their reproducing role. The conformist agent's scoring function weights $(\alpha_a, \beta_a, \gamma_a, \delta_a)$ are a formal model of habitus: fixed parameters that orient navigation toward already-dominant nodes ($\mathrm{attr}(j, G) = \mathrm{PageRank}_G(j)$, representing accumulated symbolic capital) without explicit strategic intent.

The post-consolidation graph topology — where a small set of high-weight edges concentrates navigational flow — corresponds to Bourdieu's *field*: the structured space of positions in which some trajectories are objectively more probable than others, not because they are coerced but because the structure of the field channels movement.

Foucauldian Discourse

Foucault's *archive* [Foucault, 1972] describes the system of rules governing what can be said, thought, and known at a given historical moment — not through explicit prohibition but through the structural organization of discourse. The graph after conformist consolidation instantiates this: concepts (nodes) that once had accessible paths from common entry points become unreachable not because they are forbidden but because their connecting edges have decayed below θ_{prune}. Epistemic exclusion is topological, not juridical.

A.3 Formal Intervention Framework

Given Proposition 2 — that initialization advantage is ineffective against scoring function geometry — interventions must target the model parameters rather than the agents' initial positions. Table 6 formalizes four parametric strategies, with their computational, neurobiological, and social-structural translations.

Table 6: Parametric intervention strategies for reversing reward-induced topological capture.

Strategy	**Parametric operation**	**Neurobiological instance**	**Social instance**
Structural reset	Raise $\lambda \geq 0.5$ (entropy flooding)	Extended stimulus deprivation; deep sleep induction; dopamine fast	Disruption of dominant institutional routines; forced diversity mandates
Memory saturation	Lower $\lambda \to 0$ (inhibit pruning)	Temporary synaptic pruning inhibition; extended reminiscence	Archival recovery; mandatory preservation of minority discourse; oral history programs
Reward redesign	Replace $\text{attr}(j, G)$ in f_a with entropy bonus $H(j, G)$; penalize centroid proximity	Reframe prediction error toward novelty rather than coherence (intrinsic motivation circuits)	Restructure quality metrics to reward peripheral contribution; citation diversity indexes
Temperature injection	Increase T_a (stochastic perturbation)	Break spatial/contextual behavioral routines; forced context shift	Institutional rotation; random assignment to unfamiliar roles; deliberate destabilization of habitual practice

A critical constraint from the main results applies to all interventions: strategies that operate exclusively on agent initialization (granting the weaker agent head-start advantages) are ineffective under the current scoring function geometry (Proposition 2). Durable structural redistribution requires either modifying the reward function itself (row 3) or operating outside the critical λ regime (rows 1–2). This has direct implications for diversity and inclusion policies in algorithmic systems: parity in initial conditions is insufficient if the evaluation metric is structurally biased toward dominant patterns.

Scope and Limitations of the Isomorphisms

The mappings identified above are structural, not ontological. We claim that competitive graph plasticity and the systems described share formal properties (differential reinforcement, selective decay, emergent Gini inequality, phase transitions), not that neurons, social actors, and computational agents are identical entities. The isomorphisms generate hypotheses — about, for instance, what value of cortisol corresponds to $\lambda = 0.3$ in a given cognitive task, or what institutional mechanism produces the effective decay rate in a given social field — but these hypotheses require independent empirical investigation in their respective domains.

The central contribution of the isomorphic analysis is that it identifies a single formal mechanism — reward-proportional Hebbian reinforcement under selective decay — operating across scales from synapse to society. This suggests that structural inequality in shared modifiable systems is not a historical accident or a local policy failure but a robust emergent property of a class of dynamical systems whose parameters include λ and the geometry of f_a.

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